

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## May-Nov 2020 P1HR Worked Solutions

MayNov 2020 | P1HR

27 questions   ■   question image followed by worked solution

PREPARED BY

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**PAST PAPER QUESTION****Q#: 1****Marks: 3****TOPIC: Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

1 Show that  $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

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(Total for Question 1 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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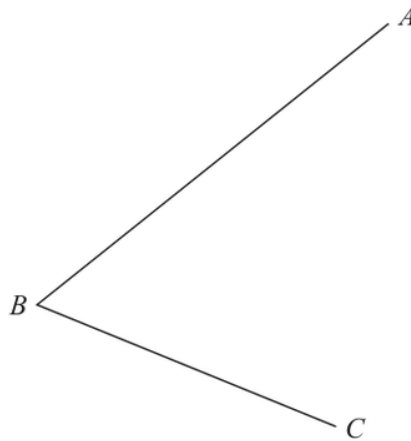
No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 2****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 2 Using ruler and compasses only, construct the bisector of angle  $ABC$ .  
You must show all your construction lines.



(Total for Question 2 is 2 marks)

## PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**3** (a) Simplify  $h^7 \times h^2$ .....  
(1)

$$G = c^2 - 4c$$

(b) Find the value of  $G$  when  $c = -5$  $G =$  .....  
(2)(c) Solve  $\frac{5x-3}{4} = 2x+3$ 

Show clear algebraic working.

 $x =$  .....  
(3)**(Total for Question 3 is 6 marks)**

## PAST PAPER SOLUTION

Q#: 3

Marks: 6

TOPIC: **Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 4****Marks: 5****TOPIC: Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 4 The table gives information about the length of time, in minutes, that each of 60 students took to travel to school on Monday.

| Length of time<br>( $t$ minutes) | Frequency |
|----------------------------------|-----------|
| $0 < t \leq 10$                  | 4         |
| $10 < t \leq 20$                 | 10        |
| $20 < t \leq 30$                 | 15        |
| $30 < t \leq 40$                 | 25        |
| $40 < t \leq 50$                 | 6         |

- (a) Write down the modal class interval.

.....  
(1)

- (b) Work out an estimate for the mean length of time taken by these 60 students to travel to school on Monday.  
Give your answer correct to one decimal place.

..... minutes  
(4)

(Total for Question 4 is 5 marks)



## PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 5****Marks: 6****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 5** In 2017, the population of a village was 7500  
In 2019, the population of the village was 8265

(a) Work out the percentage increase in the population of the village from 2017 to 2019

..... %  
(3)

In a sale, normal prices are reduced by 30%  
The sale price of a T-shirt was 31.50 euros.

(b) Work out the normal price of the T-shirt.

..... euros  
(3)

(Total for Question 5 is 6 marks)

## PAST PAPER SOLUTION

Q#: 5

Marks: 6

TOPIC: Percentages

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 6** **Marks: 2****TOPIC: Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 6 Point  $A$  has coordinates  $(-3, 11)$   
Point  $B$  has coordinates  $(47, b)$   
The midpoint of  $AB$  has coordinates  $(a, -19)$   
Find the value of  $a$  and the value of  $b$ .

 $a = \dots\dots\dots$  $b = \dots\dots\dots$ 

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**(Total for Question 6 is 2 marks)**

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EA

## PAST PAPER SOLUTION

Q#: 6

Marks: 2

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 7 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.  
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.  
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

(Total for Question 7 is 4 marks)

EA

## PAST PAPER SOLUTION

Q#: 7

Marks: 4

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 8** Harold bought an antique clock for £1200  
The clock increased in value by 8% per year.

Find the value of the clock exactly 3 years after Harold bought the clock.  
Give your answer correct to the nearest £.

£ .....

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(Total for Question 8 is 3 marks)

EA



## PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 9** A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.

The pressure on the table due to the box is 34.8 newtons/m<sup>2</sup>

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

**(Total for Question 9 is 3 marks)**

EA

## PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 10****Marks: 5****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**10** Alex makes 80 cakes to sell.

He makes chocolate cakes, lemon cakes and fruit cakes where

$$\begin{array}{ccccc} \text{number of} & : & \text{number of} & : & \text{number of} \\ \text{chocolate cakes} & & \text{lemon cakes} & & \text{fruit cakes} \end{array} = 3 : 2 : 5$$

Alex sells

all of the chocolate cakes

 $\frac{3}{4}$  of the lemon cakes $\frac{7}{8}$  of the fruit cakes

The profit he makes on each cake he sells is shown in the table.

| Type of cake | Profit per cake he sells |
|--------------|--------------------------|
| chocolate    | £2.00                    |
| lemon        | £1.70                    |
| fruit        | £2.40                    |

Work out the total profit that Alex makes from the cakes he sells.

£.....

**(Total for Question 10 is 5 marks)**

## PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: Percentages

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

11 The frequency table gives information about the ages of the 80 people in a train carriage.

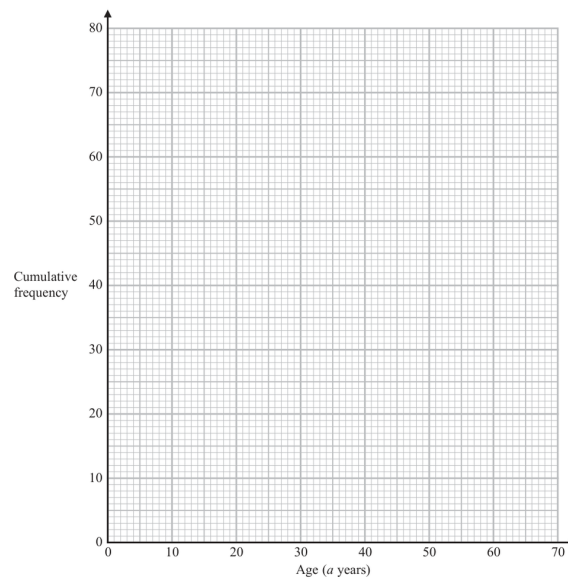
| Age ( $a$ years) | Frequency |
|------------------|-----------|
| $0 < a \leq 20$  | 9         |
| $20 < a \leq 30$ | 19        |
| $30 < a \leq 40$ | 17        |
| $40 < a \leq 50$ | 18        |
| $50 < a \leq 60$ | 13        |
| $60 < a \leq 70$ | 4         |

(a) Complete the cumulative frequency table.

| Age ( $a$ years) | Cumulative frequency |
|------------------|----------------------|
| $0 < a \leq 20$  |                      |
| $0 < a \leq 30$  |                      |
| $0 < a \leq 40$  |                      |
| $0 < a \leq 50$  |                      |
| $0 < a \leq 60$  |                      |
| $0 < a \leq 70$  |                      |

(1)

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median age of the people in the train carriage.

\_\_\_\_\_ years  
(2)

(Total for Question 11 is 5 marks)

## PAST PAPER SOLUTION

Q#: 11

Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**12** Solve the simultaneous equations

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

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**(Total for Question 12 is 4 marks)**

EA



## PAST PAPER SOLUTION

Q#: 12

Marks: 4

TOPIC: **Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 13****Marks: 6****TOPIC: Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**13** The curve **C** has equation  $y = 5x^3 - x^2 - 6x + 4$ (a) Find  $\frac{dy}{dx}$ 

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

There are two points on the curve **C** at which the gradient of the curve is 2(b) Find the  $x$  coordinate of each of these two points.  
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

**(Total for Question 13 is 6 marks)**

## PAST PAPER SOLUTION

Q#: 13

Marks: 6

TOPIC: **Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 14****Marks: 3****TOPIC: Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**14** Expand and simplify  $(4x + 1)(x - 3)(5x + 6)$ 

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**(Total for Question 14 is 3 marks)**

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EA

## PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

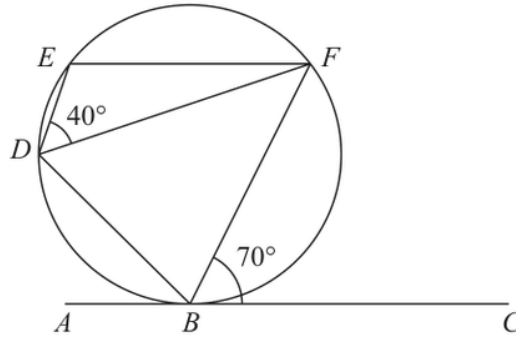
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**15**Diagram **NOT**  
accurately drawn

$B, D, E$  and  $F$  are points on a circle.  
 $ABC$  is the tangent to the circle at  $B$ .

Angle  $EDF = 40^\circ$

Angle  $FBC = 70^\circ$

Prove that the tangent  $ABC$  is parallel to  $EF$ .  
 Give a reason for each stage of your working.

(Total for Question 15 is 4 marks)

## PAST PAPER SOLUTION

Q#: 15

Marks: 4

TOPIC: **Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 16****Marks: 9****TOPIC: Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**16** The functions  $f$  and  $g$  are defined as

$$f: x \mapsto 5x - 7$$

$$g: x \mapsto \frac{5x}{x+4}$$

(a) Write down the value of  $x$  that must be excluded from any domain of  $g$ .....  
(1)(b) Find  $gf(2.6)$ .....  
(2)(c) Solve  $fg(x) = 2$  $x =$  .....  
(3)(d) Express the inverse function  $g^{-1}$  in the form  $g^{-1}: x \mapsto \dots$  $g^{-1}: x \mapsto$  .....  
(3)**(Total for Question 16 is 9 marks)**



## PAST PAPER SOLUTION

Q#: 16

Marks: 9

TOPIC: **Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 17****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 17 The diagram shows a prism  $ABCDEFGH$  with a horizontal base.

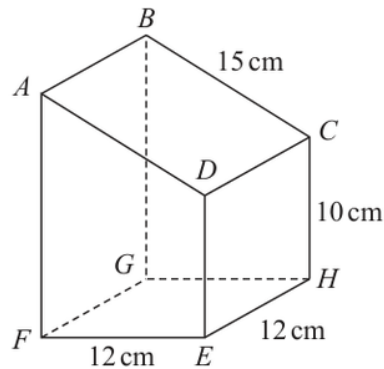


Diagram **NOT**  
accurately drawn

The base of the prism,  $EFGH$ , is a square of side 12 cm.

Trapezium  $ADEF$  is a cross section of the prism where  $AF$  and  $DE$  are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between  $CF$  and the base  $EFGH$ .  
Give your answer correct to one decimal place.

.....  
(3)

- (b) Work out the length of  $BE$ .  
Give your answer correct to one decimal place.

..... cm  
(3)

(Total for Question 17 is 6 marks)

## PAST PAPER SOLUTION

Q#: 17

Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 18****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 18** In an arithmetic series, the 6th term is 39  
In the same arithmetic series, the 19th term is 7.8

Work out the sum of the first 25 terms of the arithmetic series.

(Total for Question 18 is 4 marks)

EA

## PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 19** The diagram shows rectangle  $ABCD$  with rectangle  $EFGH$  cut out to form the shaded region.

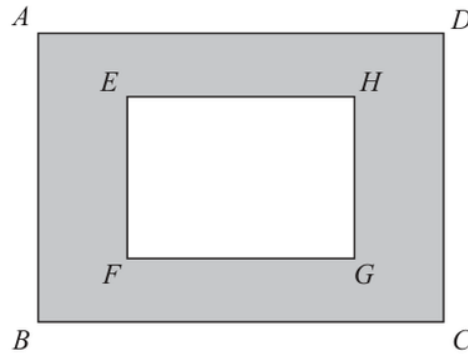


Diagram **NOT**  
accurately drawn

$AD = 8.3$  cm correct to one decimal place

$DC = 7.2$  cm correct to one decimal place

$EH = 6.2$  cm correct to one decimal place

$HG = 5.3$  cm correct to one decimal place

Work out the upper bound of the area of the shaded region.  
Show your working clearly.

.....  $\text{cm}^2$

(Total for Question 19 is 3 marks)

## PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**20** A curve has equation  $y = f(x)$ 

There is only one maximum point on the curve.

The coordinates of this maximum point are  $(-3, 4)$ 

Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x) - 6$

(....., .....)

(ii)  $y = f(2x)$

(....., .....)

**(Total for Question 20 is 2 marks)**

EA



## PAST PAPER SOLUTION

Q#: 20

Marks: 2

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**20** A curve has equation  $y = f(x)$ 

There is only one maximum point on the curve.

The coordinates of this maximum point are  $(-3, 4)$ 

Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x) - 6$

(....., .....)

(ii)  $y = f(2x)$

(....., .....)

**(Total for Question 20 is 2 marks)**

EA

## PAST PAPER SOLUTION

Q#: 20

Marks: 2

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21** **Marks: 5****TOPIC:** Algebraic Roots & Indices

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**21** Given that  $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of  $n$  for which  $M = 2$

(Total for Question 21 is 5 marks)

EA

## PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**21** Given that  $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of  $n$  for which  $M = 2$

(Total for Question 21 is 5 marks)

EA

## PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

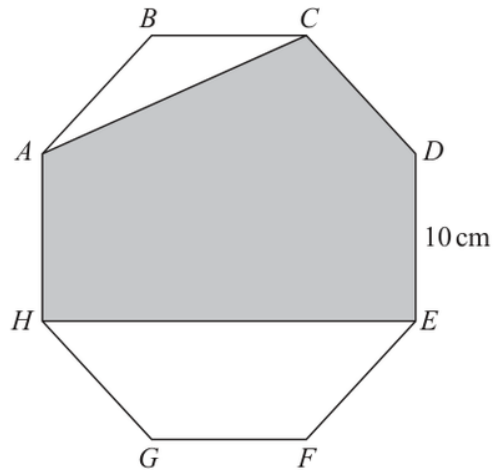
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**22** The diagram shows a regular octagon  $ABCDEFGH$ .Diagram **NOT**  
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region  $ACDEH$ .Give your answer correct to the nearest  $\text{cm}^2$ .....  $\text{cm}^2$ **(Total for Question 22 is 6 marks)**



## PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

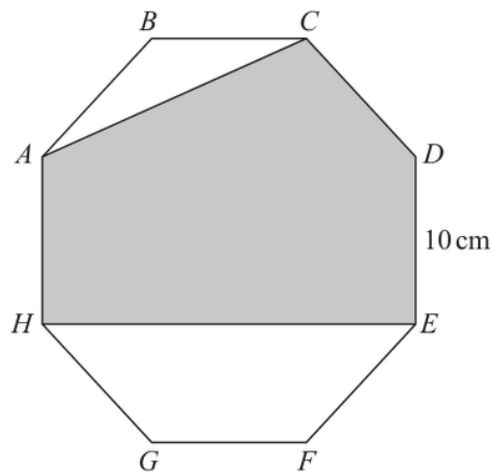
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**22** The diagram shows a regular octagon  $ABCDEFGH$ .Diagram **NOT**  
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region  $ACDEH$ .Give your answer correct to the nearest  $\text{cm}^2$ .....  $\text{cm}^2$ **(Total for Question 22 is 6 marks)**

## PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 23****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**23** In a bag, there are only

3 blue beads  
4 white beads  
and  $x$  orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is  $\frac{3}{8}$

Find the total number of beads in the bag.  
Show clear algebraic working.

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(Total for Question 23 is 4 marks)

EA

## PAST PAPER SOLUTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 23****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

**23** In a bag, there are only

3 blue beads  
4 white beads  
and  $x$  orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is  $\frac{3}{8}$ 

Find the total number of beads in the bag.  
Show clear algebraic working.

---

(Total for Question 23 is 4 marks)

EA

## PAST PAPER SOLUTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA